

# Session 11

Guiding principles and best practices for data capacity building

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# Building data capacity

- Not everyone needs to be a data scientist.
- Data capacities in policymaking refer to how well infrastructure, systems, and people can use data effectively for decisions.
- Strengthening these abilities across different teams is essential.
- Complementary skill sets help reinforce the entire data value chain and improve overall decision-making.

# 1. Identify gaps in data skills and statistical capacities

# Types of skills



## Technical know-how

Mechanical, ICT, mathematical, scientific and/or other technology-related skills such as proficiency with a standard statistical package (e.g., R; Stata; SPSS) to carry out data processing, analysis and dissemination.



## Work know-how

An employee's knowledge and understanding of work practices, processes, statistical concepts, definitions and classifications, business models, and organisational culture in use in their organisation, as well as about the power relationships within and between organisations.



## Problem-solving and creative thinking

The traits that support defining, approaching and analysing a problem in a technically correct manner and from a fresh perspective to devise new ways for carrying out tasks, meeting challenges and solving problems.

# Data Capability Framework

|                                                             | PLAN                                | COLLECT                             | DESCRIBE                            | STORE                               | ANALYSE                             | USE                                 | SAVE/DESTROY                        |
|-------------------------------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Employ data coding and classification principles            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Integrate data                                              | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Use data processing methodologies                           | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Contribute to data outputs, products or service production  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Perform exploratory data analysis                           | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Conduct business intelligence data analysis                 | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Conduct statistical data analysis                           | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Conduct specialist data analysis                            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Identify and understand data availability                   | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Employ data collection methodology                          | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Contribute to data access design                            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Contribute to the sourcing and use of administrative data   | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Understand and contribute to data collection process design | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Describe and summarise data                                 | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Understand and apply data editing methods                   | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Use data quality assurance measures                         | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Identify and evaluate data intelligence                     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Improve data processes/systems/products                     | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Identify research questions                                 | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Apply data governance guidance                              | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Value organisational data as assets                         | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Employ statistical concepts and methodologies               | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Enable others to use and re-use data                        | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Employ data and information management concepts             | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| Visualise data                                              | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |

# Data Capability Framework - example

- **Capability:** Employ data coding and classification principles
- **Category:** Describe, Store, Use, and Save/Destroy

## New

- Is aware of relevant data classifications and coding protocols, and their proper application to data in general.
- Knows who to consult for expert knowledge.

## Proficient

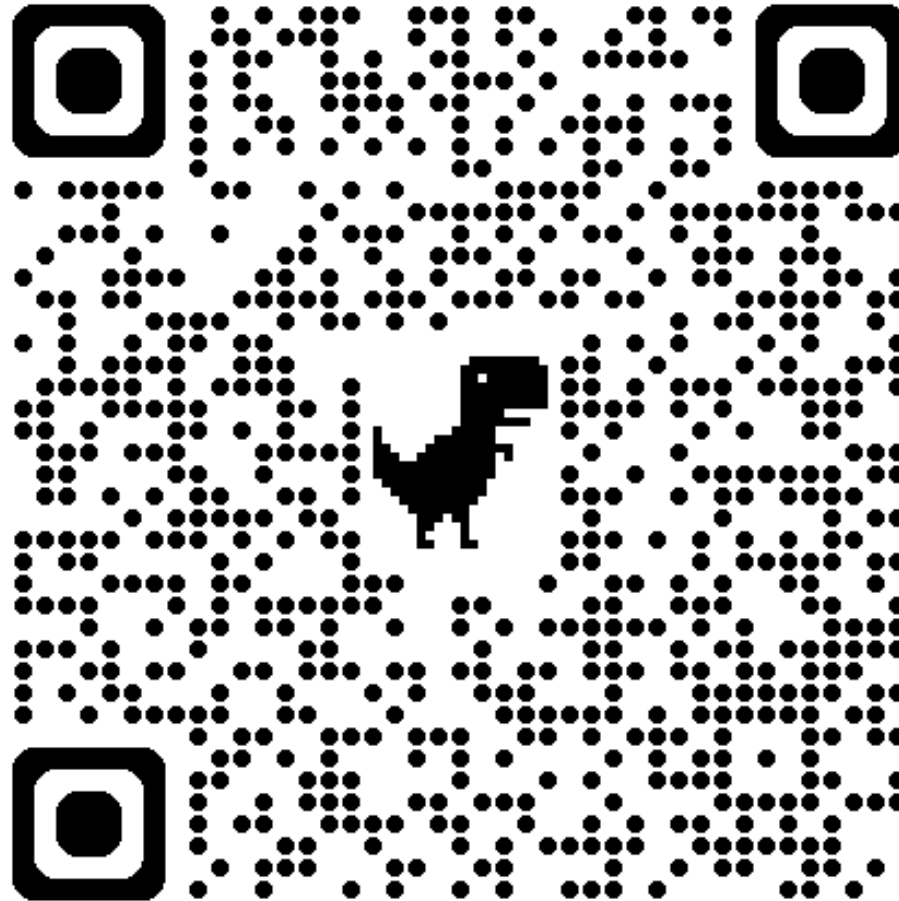
- Has a comprehensive knowledge of data classifications and coding protocols.
- Knows where to obtain expert advice about coding and classifications as needed.

## Expert

- Is consulted regularly by others about data classifications and coding protocols.
- Can employ conceptual frameworks in support of data classification and coding.



# Data Capability Framework Guide



Link



# Data Maturity Assessment Framework

**Topic:** Engaging with others

**Topic:** Managing and using data ethically

**Topic:** Having the right data skills and knowledge

**Topic:** Managing your data

**Topic:** Having the right systems

**Topic:** Protecting your data

**Topic:** Knowing the data you have

**Topic:** Setting your data direction

**Topic:** Making decisions with data

**Topic:** Taking responsibility for data

# Data Maturity Assessment Framework

- **Themes:** Uses, Data, Leadership, Culture, Tools, Skills
- **Maturity Levels:** Beginning, Emerging, Learning, Developing, Mastering

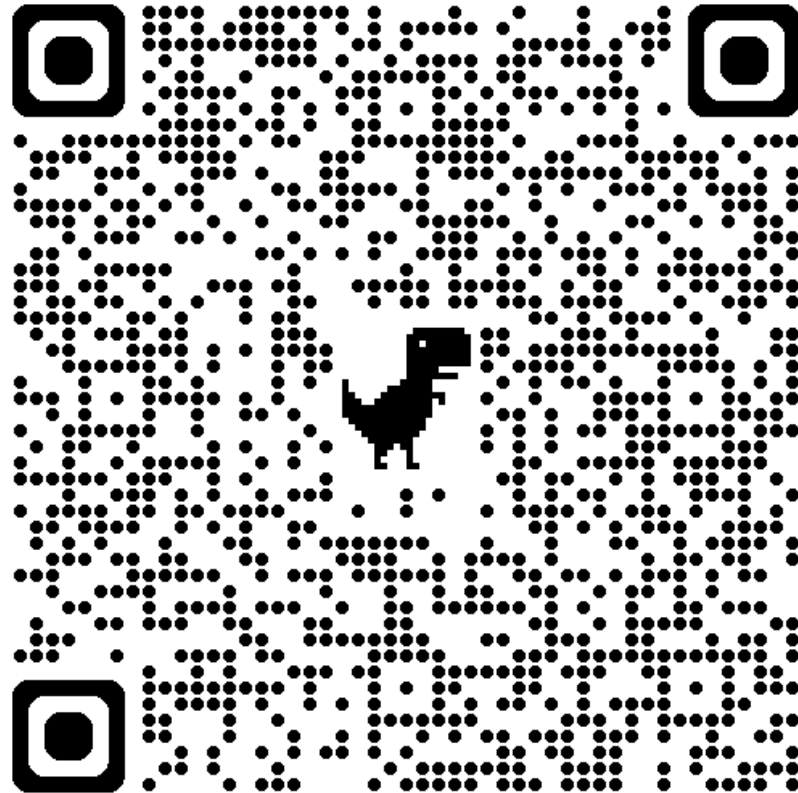
# Data Maturity Assessment Framework

## - example

**Topic:** Engaging with others

| ID | Row summary                                 | Theme          | Level 1: Beginning                                                                                               | Level 2: Emerging                                                                                                                                       | Level 3: Learning                                                                                                                                                                                                                                                                           | Level 4: Developing                                                                                                                                                                                                                                | Level 5: Mastering                                                                                                                                                                        |
|----|---------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Making data available to those who need it. | <b>Culture</b> | Makes data available by default only to a single specialist person or team. Discourages sharing data internally. | Beginning to share data internally either verbally or via reports. Does not encourage data sharing across teams or provide ways to share data directly. | Enables specialist users to access and share some data internally, but systems or processes limit access to and sharing of data by default rather than by design. Some internal users have appropriate access to data they need but may require specialist support to access or share data. | Beginning to provide ways to access and share data directly, but non-expert staff may require some intervention from specialists to do so. All internal users and some external users have appropriate access to data they need when they need it. | Data can be accessed and directly shared appropriately by all users who need it. All internal and external users can access data they need when they need it, without specialist support. |

# Data Maturity Assessment Framework Guide

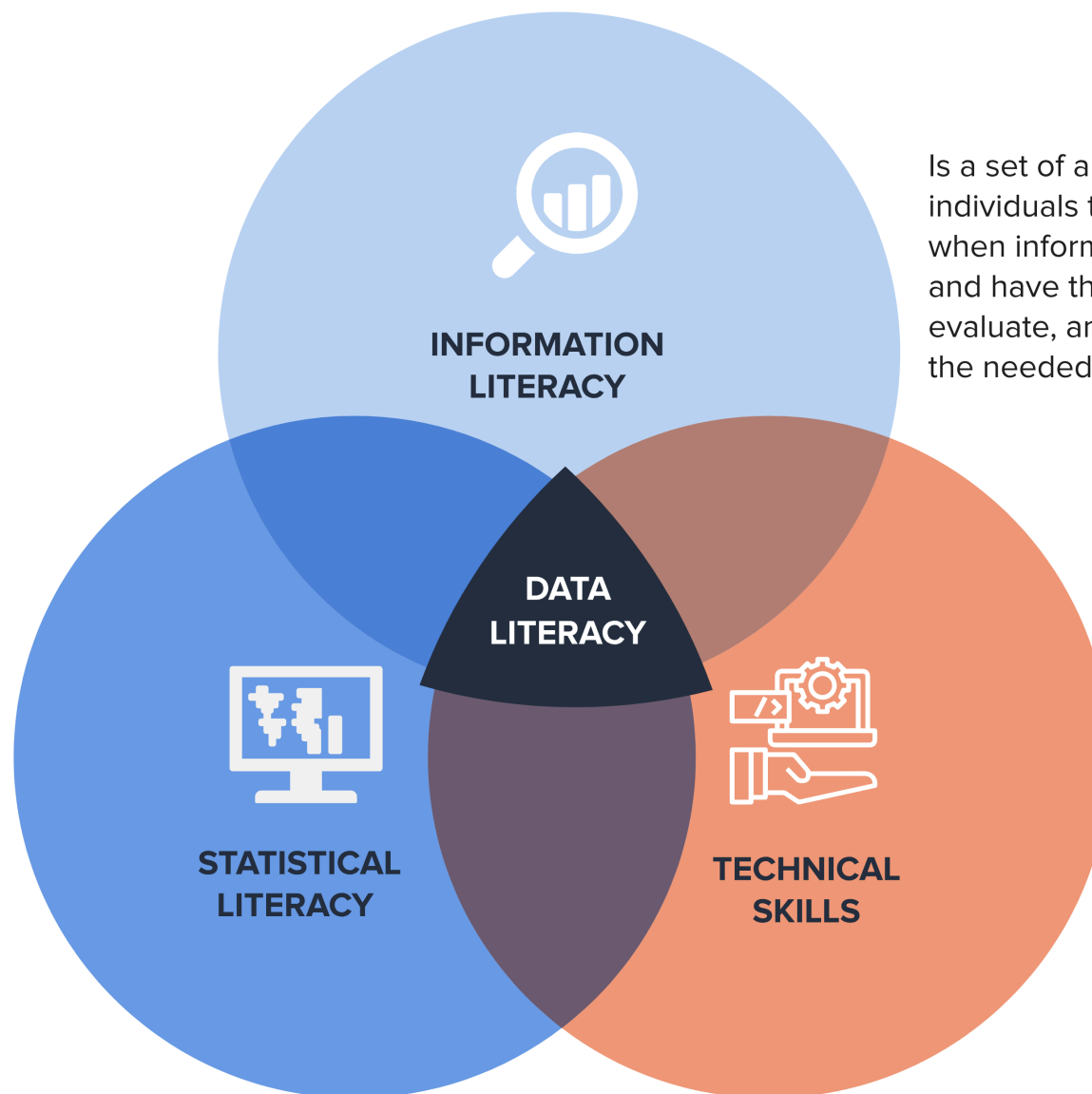


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## 2. Build data capacities in your team

# Data Literacy

Refers to use of statistics as evidence, encompassing what to count or measure, how to assemble those measurements into summary statistics, what comparisons to form from these statistics and how to communicate these statistics.



Is a set of abilities requiring individuals to “recognize when information is needed and have the ability to locate, evaluate, and use effectively the needed information.

Refer to skills that are essential for handling, analyzing, interpreting, and deriving insights from data, which has become increasingly important in the digital age.

Source: UN Data Revolution Group

# Data Skills Framework



# 3. Operationalise data capacity building



# Short-term

- Training workshops
- Online training resources
- Knowledge sharing on global innovations and use cases

# Medium-term

- Internal data champions
- Periodic data skills training
- Pilot data projects
- Build complementary skills with other stakeholders
- Build on local knowledge and capacities
- Build network of data practitioners across your organization

# Long-term

- Embed data skills in job roles
- Institutionalize data training programs
- Partner with think tanks and external organizations
- Partner with academia and local universities